AVL Trees

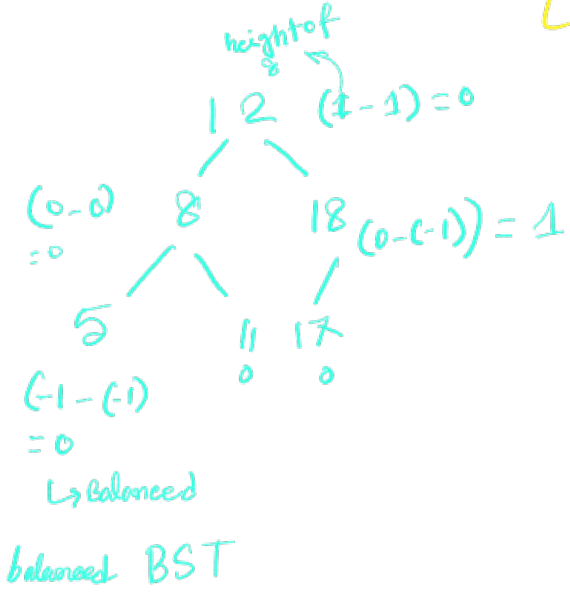
BST → O(h) ~ log(n)

- Self balancing.
- balance factor = height of left subtree − " right "
- balance " = {-1, 0, 1} => tree is balanced

When the tree will be imbalanced

↳ Balance using Rotation.

Codecrafters. C++ OOP job → DB



left-heavy : bf > 0

right " : bf < 0

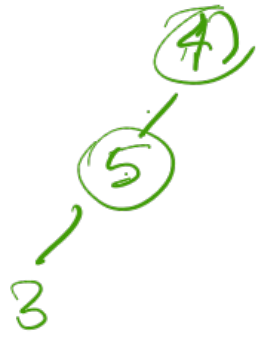
look for imbalanced notation from bottom to top

fix the 1st imbalanced node (all others auto balanced).

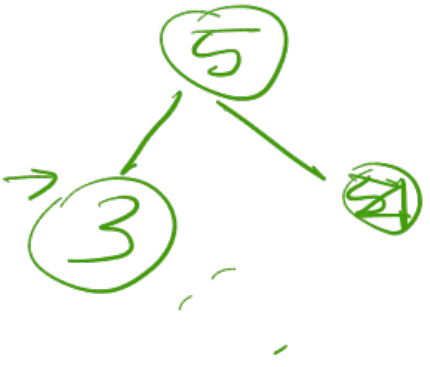
AVL Trees

balance factor → positive (heavy on left)

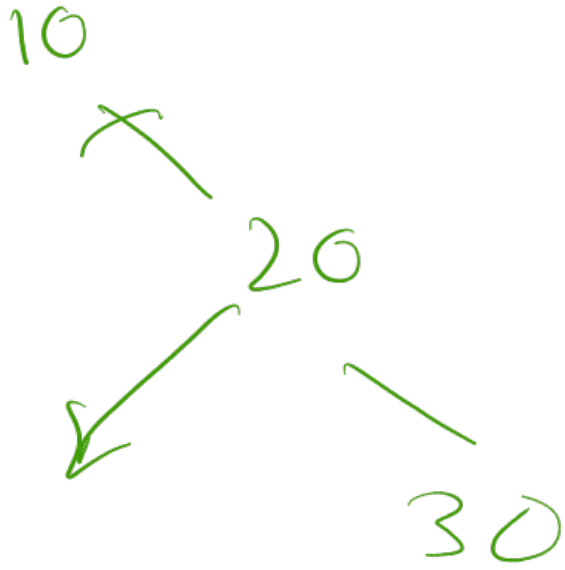
Max height ~ 1.44 log N.



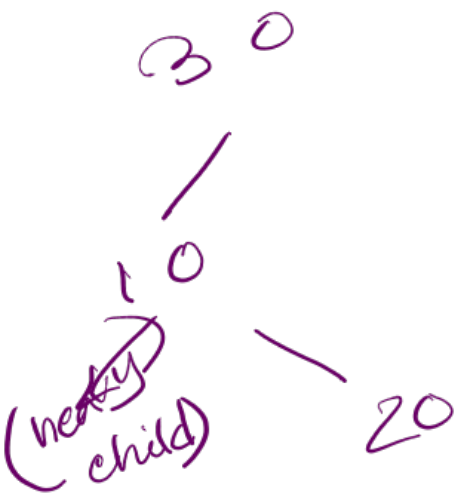
right rot →



RR case (Right rotate)



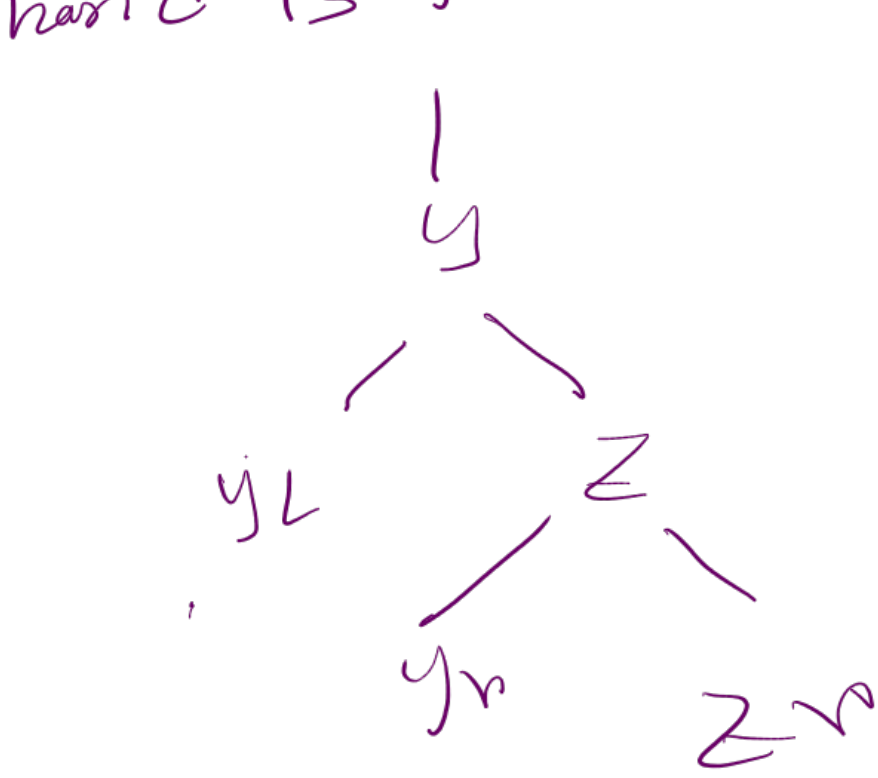
LR → convert LL → (left rotate heavy child) → RR

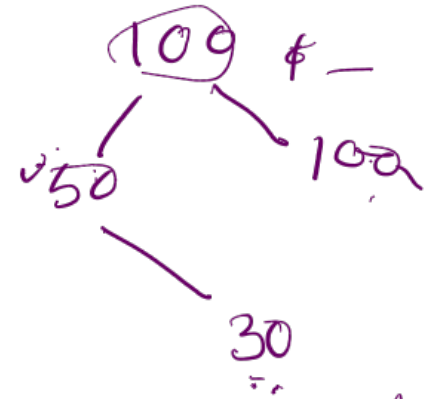


heavy child (child of imbalanced node)

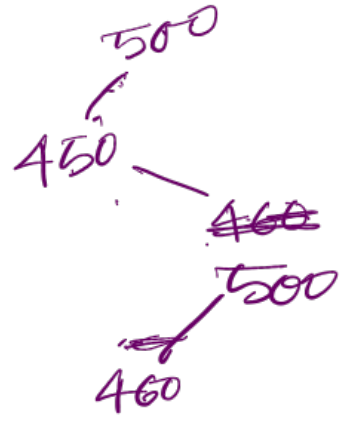
RR → RR →

Y.L → goes to some whose right





LR → 2 rotation → 1st heavy child then imbalanced node.



node* z
" * y = z.Right